

GroupMate — Day 3 Progress Report

Building AI Application Challenge 2026 | Rana Hamad | LLM/API Integration Path | 2026-09-09

Application Concept

GroupMate is an AI-powered task distribution app that fixes the most painful part of student group projects: splitting work fairly. Members pick their strong suits from AI-generated, project-specific skill tags; the AI then distributes tasks by skill fit and workload balance, and the app tracks progress, flags anyone falling behind, and keeps the group aligned in one place.

API Enhancements (LLM/API Integration Path)

- Two core **OpenAI (gpt-4o)** integrations in JSON response mode: **generate_tags()** produces project-specific skill tags from the project description, and **ai_distribute_tasks()** assigns tasks fairly by skill match and balanced workload, including logic to mark large tasks as shared.
- Added a retry wrapper (**_call_openai_json**) so an occasional malformed JSON response is retried automatically instead of failing the request.
- Added a third AI integration: an in-app **Ask AI** assistant (context-aware chat) that answers teamwork, scheduling, and conflict-resolution questions using the group's own project description as context.
- Full Supabase backend wired in end-to-end (groups, members, tasks, files, messages, help_requests) and tested live against real OpenAI and Supabase credentials, not just locally mocked.
- Built out the full product surface in Gradio: an onboarding wizard (create/join group, add members, submit strong suits) leading into a persistent sidebar dashboard with live SVG progress rings, group progress tracking, group + private chat, file sharing with Supabase Storage, and a Request/Offer Help flow.

Challenges Encountered & How They Were Addressed

- **Keeping AI JSON output reliably parseable** — added automatic retry logic around every OpenAI JSON call rather than trusting a single attempt.
- **Local environment quirks** (Windows' python alias, no GitHub CLI installed) — resolved via the **py** launcher and a manual repo-create-and-push workflow.
- **OpenAI billing had to be set up before any live testing** — added credit, then ran a full live smoke test (create group → AI tags → add members → strong suits → AI task distribution → dashboard → tasks) confirming correct, fair output.
- **UI direction evolved through several iterations** (theme colors, a step wizard, then a persistent sidebar shell matching a hand-drawn wireframe) — kept backend logic and UI cleanly separated from the start, so each redesign only touched the interface layer, not the data model or AI logic.

Current Status & Plans for Adjustment

The core MVP loop is complete and verified live: group creation with AI tag generation, member management, strong-suit submission, automatic AI task distribution, a progress dashboard with a falling-behind flag, file sharing, group/private chat, a help-request flow, and an AI assistant — all tested against real OpenAI and Supabase accounts. Remaining before final submission: Hugging Face Spaces deployment, a demo video, the LinkedIn post, and running the prepared 12-sample evaluation set to benchmark task-split fairness across project types.

Repo: github.com/ranasalahe/GroupMate (private — access will be shared with judges/mentors as needed).